SSAT · MATH · PAPER EDITION

Every math question you've missed

137 real questions from your timed Quant sections since September (Mock 7 to Mock 10.5c), sorted by topic and split into 7 days.

WARREN — HOW TO DO IT ON PAPER

1. Do **one day** per sitting. Write the date at the top of the day before you start.
2. Work in the blank space under each question, like on the real test. Show your steps.
3. **Circle one letter** for each answer. Your first answer is the one that counts: no erasing after you check.
4. ⚠ **missed twice** means you got it wrong on the test *and* on the retry. Slow down on those.
5. Each topic starts with a grey box of reminders. Read it before the questions.
6. Not timed, but write down how many minutes the day took. Then hand the pages in to be checked.

DAY	QUESTIONS	TOPICS	DATE	MINUTES	SCORE
Day 1	1–20	Percent problems; Fractions & decimals			/ 20
Day 2	21–40	Ratios, rates & units; Algebra: equations & expressions			/ 20
Day 3	41–60	Algebra: equations & expressions; Number sense & arithmetic			/ 20
Day 4	61–80	Number sense & arithmetic; Multi-step word problems			/ 20
Day 5	81–99	Angles, triangles & shapes; Perimeter, area & volume			/ 19
Day 6	100–118	Perimeter, area & volume; Charts, averages & probability			/ 19
Day 7	119–137	Charts, averages & probability; Number properties & logic			/ 19

Day 1 · Questions 1–20

20 questions

Date: _____ Minutes: _____ Score: _____ / 20

% Percent problems

- Ask "percent **of what**?" — the word after *of* or *than* is the whole.
- Percent change = change ÷ the number you **started** from.
- Tax and tip are both on the base: total = base × 1.30, so divide.

1. If a harvest yields 60 bushels of corn, 100 bushels of wheat, and 80 bushels of soybeans, what percent of the total harvest is corn?

(A) 25 % (B) 30 % (C) 33 % (D) 40 % (E) 50 %

2. A sportswriter claims that her football predictions are accurate 40% of the time. During football season, a fan kept records and found that the writer was inaccurate for a total of 30 games, although she did maintain her 40% accuracy. For how many games was the sports writer accurate? ⚠ MISSED TWICE

(A) 5 (B) 15 (C) 20 (D) 40 (E) 60

3. Dinner (plus tax and tip) cost $ 145.6. The tax rate is 10 % and Mr. Simmons left a 20 % tip. Both tax and tip are calculated on the base amount of the check. What was the base amount of Mr. Simmons's bill? ⚠ MISSED TWICE

(A) $ 78.00 (B) $ 113.32 (C) $ 77.41 (D) $ 112.00 (E) $ 81.30

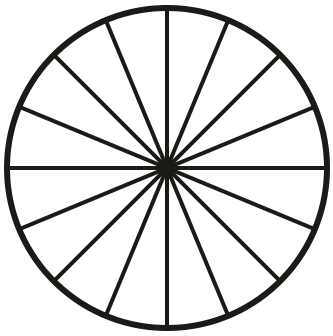
4. The Price of a jacket is reduced by half, and the resulting price is then reduced by 10%. The final price is what percentage of the original price?

(A) 10 % (B) 40 % (C) 45 % (D) 55 % (E) 60 %

5. How much larger than 80 is 100 _____.

(A) 18 % (B) 20 % (C) 25 % (D) 35 % (E) 40 %

6. If Harry eats 4 pieces of the pizza represented below, what percent of the pizza does he consume?



(A) 4% (B) 25% (C) 30% (D) 40% (E) 55%

7. Of the following fractions, which is closest to 37%?

(A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{2}{5}$ (D) $\frac{3}{7}$ (E) $\frac{3}{8}$

8. Chaim has $4,500 in a savings account. If last year he had $5,000 in the savings account, by what percent did the money in the account decrease from last year to now?

(A) 11% (B) 10% (C) 9% (D) 5% (E) 1%

½ Fractions & decimals

- Turn mixed numbers into improper fractions before you multiply or divide.
- Line decimals up by the point, then compare one place at a time.
- "Of what was left" → multiply by what is left, not by the whole.

9. Sophie is making bows for packages and uses 1.25 m of ribbon per bow. If she has 12 m and makes as many bows as possible, how much ribbon will she have left?

(A) 0.6 m (B) 2.4 m (C) 0.72 m (D) 0.7 m (E) 0.75 m

10. What is the value of the sum of $\frac{1}{3}$ and $\frac{2}{5}$?

(A) $\frac{3}{8}$ (B) $\frac{11}{15}$ (C) $\frac{11}{30}$ (D) $\frac{4}{5}$ (E) $\frac{1}{4}$

11. How will $\frac{4}{5}$ be written as a percent?

(A) 40% (B) 125% (C) 90% (D) 80% (E) 50%

12. Arrange the following numbers from least to greatest value: 0.85, $\frac{4}{5}$, $\frac{2}{3}$, $\frac{9}{10}$, $\frac{91}{100}$

(A) 0.85, $\frac{4}{5}$, $\frac{2}{3}$, $\frac{9}{10}$, $\frac{91}{100}$

(B) $\frac{4}{5}$, 0.85, $\frac{2}{3}$, $\frac{9}{10}$, $\frac{91}{100}$

(C) $\frac{2}{3}$, $\frac{4}{5}$, 0.85, $\frac{9}{10}$, $\frac{91}{100}$

(D) 0.85, $\frac{4}{5}$, $\frac{2}{3}$, $\frac{91}{100}$, $\frac{9}{10}$

(E) $\frac{2}{3}$, $\frac{4}{5}$, $\frac{9}{10}$, 0.85, $\frac{91}{100}$

13. $12\frac{1}{2} \div \frac{1}{2} + \frac{1}{2} \times 12 - 3 =$ ⚠ MISSED TWICE

(A) 1 (B) $4\frac{3}{4}$ (C) 20 (D) 28 (E) $32\frac{1}{2}$

14. Evaluate: $\dfrac{1\frac{3}{4} - \frac{1}{4}}{\frac{1}{8}}$ _____.

(A) 1 (B) 2 (C) 12 (D) 13 (E) 14

15. On Monday, Gerri ate $\frac{1}{8}$ of an apple pie. On Tuesday, she ate $\frac{1}{4}$ of what was left of the pie. What fraction of the entire pie did Gerri eat on both days?

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{5}{8}$ (D) $\frac{7}{8}$ (E) $\frac{11}{32}$

16. Which of the following fractions is not more than one third? ⚠ MISSED TWICE

(A) $\frac{22}{63}$ (B) $\frac{4}{11}$ (C) $\frac{15}{46}$ (D) $\frac{33}{98}$ (E) $\frac{102}{303}$

17. If $\frac{5}{x}$ is subtracted from $\frac{3}{x}$, the result is _____.

(A) 2 (B) $\frac{8}{x}$ (C) $-\frac{2}{x}$ (D) $\frac{2}{x}$ (E) $\frac{2}{x^2}$

18. Which fraction lies between $\frac{2}{3}$ and $\frac{4}{5}$? ⚠ MISSED TWICE

(A) $\frac{5}{6}$ (B) $\frac{17}{20}$ (C) $\frac{7}{10}$ (D) $\frac{13}{15}$ (E) $\frac{9}{10}$

19. On Monday, Aidan reads $\frac{1}{3}$ of a book, and on Tuesday, Aidan reads $\frac{1}{4}$ of the remaining pages. To finish the book, he must read an additional 60 pages. How many pages are in the book?

(A) 720 (B) 360 (C) 144 (D) 120 (E) 72

20. What is the value of $\frac{2}{3} \times \frac{1}{8} \div \frac{3}{4}$?

(A) $\frac{1}{18}$ (B) $\frac{1}{16}$ (C) $\frac{1}{12}$ (D) $\frac{1}{9}$ (E) $\frac{1}{6}$

Day 2 · Questions 21–40

20 questions

Date: _____ Minutes: _____ Score: _____ / 20

⚖️ Ratios, rates & units

- Add the ratio numbers to get the parts (3 : 5 → 8 parts).
- Put everything in the same unit first — minutes with minutes, hours with hours.
- Write the rate as a fraction with units so you can see which way to divide.

21. If Danny takes 48 minutes to walk 3 miles, how long should it take him to walk 5 miles maintaining the same speed?

(A) 32 min (B) 64 min (C) 80 min (D) 96 min (E) 128 min

22. If Sarah reads at an average rate of 21 pages in four nights, how long will it take her to read 140 pages?

(A) 6 nights (B) 26 nights (C) 8 nights (D) 27 nights (E) 28 nights

23. In Jim's school, there are 3 girls for every 2 boys. There are 650 students in total. Using this information, how many students are girls?

(A) 260 (B) 130 (C) 65 (D) 390 (E) 320

24. Keith's bakery had 252 customers go through its doors last week. This week, that number increased to 378. Express this increase as a percentage.

(A) 26% (B) 50% (C) 35% (D) 12% (E) 21%

25. How long should an object $6\frac{1}{2}$ feet long be drawn, if according to the scale, $\frac{1}{4}$ inch in the drawing equals 1 foot?

(A) $1\frac{3}{4}$ inches (B) $1\frac{5}{8}$ inches (C) $\frac{7}{8}$ inches
(D) $\frac{5}{8}$ inches (E) $\frac{17}{32}$ inches

26. A snapshot measures $1\frac{1}{4}$ inches by 2 inches. It is to be enlarged so that the longer dimension will be 4 inches. The length of the enlarged shorter dimension will be _____. ⚠ MISSED TWICE

(A) $2\frac{1}{2}$ inches (B) $2\frac{5}{8}$ inches (C) 3 inches
(D) $3\frac{3}{8}$ inches (E) $7\frac{3}{19}$ inches

27. 55 is to _____ as 110 is to 0.55.

(A) 0.275 (B) 0.900 (C) 4.50 (D) 9.00 (E) 22.5

28. If $\frac{3}{8}''$ on a scale drawing is equivalent to one foot at full scale, what distance on the drawing will stand for forty inches?

(A) $\frac{1}{8}$ inches (B) $\frac{7}{8}$ inches (C) $1\frac{1}{4}$ inches

(D) $2\frac{1}{3}$ inches (E) $8\frac{8}{9}$ inches

x Algebra: equations & expressions

- Stuck? Pick an easy number for the letter and test each choice.
- Do the same thing to both sides; when you multiply an inequality by a negative, flip it.
- Read the last line again: are they asking for x, or for $2x + 1$?

29. Find the value of y in the equation $y = 4x^3$ when $x = 2/3$.

(A) $9\frac{1}{3}$ (B) $1\frac{5}{27}$ (C) $9\frac{4}{11}$ (D) $8\frac{12}{25}$ (E) $12\frac{2}{3}$

30. Review the following five algebraic expressions and then select the correct answer below.

(a) $4(r + s) + t$

(b) $4r + 4s$

(c) $4rst$

(d) $4r(s + t)$

(e) $4rt + 4rs$

⚠ MISSED TWICE

(A) $a = b$ (B) $c > d$ (C) $e < a$ (D) $b > d$ (E) $d = e$

31. Hal spends $t + 4$ hours per day watching television. José spends t hours and Tony watches television $t - 2$ hours per day. Find the average number of hours of watching television for the three students.

(A) $4t + 2/3$ (B) $3t + 2/3$ (C) $5t - 1/3$ (D) $t + 12$ (E) $3t - 2/3$

32. Express the linear relationship indicated in the table.

x	1	2	3	4	5
y	1	3	5	7	9

(A) $y = x + 1$ (B) $y = 2x + 1$ (C) $y = 3x - 1$

(D) $y = 2x - 1$ (E) $y = x + 4$

33. Hector Villanueva has a 0.300 batting average (hits/times at bat). If he gets a total of 10 more hits the next 10 times at bat, his batting average will jump to 0.400. Determine how many hits he had originally. ⚠ MISSED TWICE

(A) 24 (B) 16 (C) 22 (D) 34 (E) 18

34. Given $5 > 3x \geq -2$, find a possible value for x. ⚠ MISSED TWICE

(A) 2 (B) 1 (C) −2 (D) −1 (E) 3

35. If $x = 4/5$, $y = 2/3$, and $z = \sqrt{3}$, find the value of $10x - 9y - 4z^2$.

(A) −6 (B) −10 (C) −8 (D) 4 (E) 2

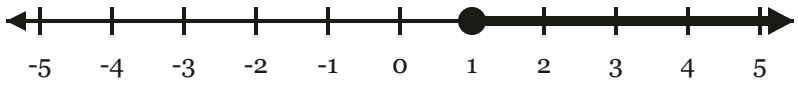
36. How will the following algebraic expression be simplified: $(5x^2 - 3x + 4) - (2x^2 - 7)$?

(A) x^5 (B) $3x^2 - 3x + 11$ (C) $3x^2 - 3x - 3$
(D) $x - 3$ (E) $5x$

37. If $-3(x + 4) = x + 8$, what is the value of x?

(A) $x = 4$ (B) $x = 2$ (C) $x = 5$ (D) $x = -5$ (E) $x = 15$

38. Which inequality represents the values displayed on the number line? ⚠ MISSED TWICE



(A) $x < 1$ (B) $x \leq 1$ (C) $x > 1$ (D) $x \geq 1$ (E) $x = 1$

39. Solve this equation: $9x + x - 7 = 16 + 2x$

(A) -4 (B) 3 (C) $\frac{9}{8}$ (D) $\frac{23}{8}$ (E) 4

40. If $4x - 3 = 5$, then $x = ?$

(A) 1 (B) 2 (C) 3 (D) 4 (E) $\frac{1}{2}$

Day 3 · Questions 41–60

20 questions

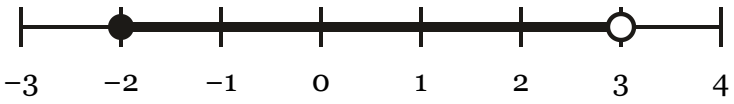
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x Algebra: equations & expressions (continued)

41. If $y + 3 > 10$, the y may be _____.

(A) larger than 7 (B) larger than 6 (C) larger than 0
(D) equal to 0 (E) unknown

42. The graph shown above is of which inequality? ⚠ MISSED TWICE



(A) $-2<x<3$ (B) $-2\le x<3$ (C) $-2\le x\le 3$ (D) $-2<x\le 3$ (E) $-2<x$

43. What is the sum of $\frac{a}{b}$ and $\frac{b}{a}$? ⚠ MISSED TWICE

(A) $\frac{a+b}{ab}$ (B) $\frac{a^2+b^2}{ab}$ (C) $\frac{(a+b)^2}{ab}$ (D) $\frac{a^2+b^2}{2ab}$ (E) 1

44. If $a = 4$ and $b = \frac{1}{5}$, then the value of a, expressed in terms if b, is _____.

(A) $25b$ (B) $20b$ (C) $5\frac{1}{5}b$ (D) $5b$ (E) $\frac{1}{25}b$

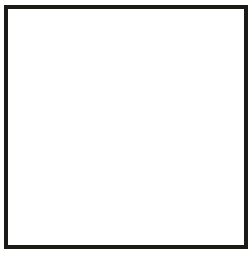
45. If $|\ 4a - 3\ | = 5$, which of the following is a possible value for a?

(A) -2 (B) -1 (C) 0 (D) 1 (E) 2

46. Bob is x years old and Jerry is 5 years older. In terms of x, what was the sum of their ages, in years, 4 years ago?

(A) $2x + 3$ (B) $2x + 2$ (C) $2x - 3$ (D) $x - 3$ (E) $x - 10$

47. The length of each side of the square above is $\frac{2x+1}{3}$. The perimeter of the square is _____.



(A) $\frac{8x}{3} + 4$ (B) $\frac{8x+4}{3}$ (C) $\frac{2x}{3} + 4$ (D) $\frac{2x}{3} + 16$ (E) $\frac{4x}{3} + 2$

48. $(3x^2y^{-4}z)^4x^3y =$ _____.

(A) $3x^{11}y^{-15}z^4$ (B) $81x^{-1}yz^4$ (C) $81x^{11}y^{-15}z^4$
(D) $3x^9yz^4$ (E) $3x^3y^{-5}z$

49. At a fund-raiser, 500 people each donated y dollars. In terms of y, what was the total number of dollars donated?

(A) 500 (B) $500y$ (C) $\frac{y}{500}$ (D) $\frac{500}{y}$ (E) $500 + y$

50. The expression $(3K^3)^2$ is equivalent to _____.

(A) $9K^6$ (B) $27K^6$ (C) $27K^5$ (D) $9K^5$ (E) $3K^5$

51. What is the x-intercept of the line described by the equation $y = 4x + 5$? ⚠ MISSED TWICE

(A) 5 (B) -5 (C) $-\frac{5}{4}$ (D) $-\frac{4}{5}$ (E) 0

52. If $a = 16$, $b = 2$, and $c = 3$, the value of $\sqrt{a + 3b + c}$ is _____.

(A) 16 (B) 7 (C) 5 (D) 4 (E) 2

53. If the length of a rectangle is 5 more than the width, w, what is the area of the rectangle in terms of w?

(A) $\frac{1}{5}w(w)$ (B) $5w(w)$ (C) $(w - 5)w$ (D) $(w + 5)w$ (E) $6w$

54. If $3 + 7 \times n = 6$, what is the value of n?

(A) -4 (B) $\frac{3}{7}$ (C) $\frac{3}{5}$ (D) $\frac{9}{7}$ (E) 2

🔢 Number sense & arithmetic

> - Name the place value before you compute anything.
> - × and ÷ before + and −; keep the minus signs.
> - Estimate first — it throws out the silly choices.

55. Simplify $6|-3| - 4|5|$. ⚠ MISSED TWICE

(A) 3 (B) 2 (C) -3 (D) -2 (E) 4

56. Find the value of $3^{\frac{9}{4}} \times 3^{7}$.

(A) 3 (B) $3\frac{1}{2}$ (C) $2\frac{1}{4}$ (D) $2\frac{3}{4}$ (E) $3\frac{3}{4}$

57. Of the following expressions, which is the smallest?

(A) $5 \times 3 - (2 + 7)$

(B) $5(3) - 48/6$

(C) $14 - 5 + 30/6$

(D) $18/2 - 3(4 - 6) + 54/9$

(E) $5(6 + 3) - 3^2$

58. How will the following number be written in standard form: $(1 \times 10^4) + (3 \times 10^3) + (7 \times 10^1) + (8 \times 10^0)$?

(A) 137 (B) 13,078 (C) 1,378 (D) 8,731 (E) 87,031

59. How will the number 847.89632 be written if rounded to the nearest hundredth?

(A) 847.90 (B) 900 (C) 847.89 (D) 847.896 (E) 847.8963

60. What is the value of the expression: $7^2 - 3 \times (4 + 2) + 15 \div 5$?

(A) 12.2 (B) 40.2 (C) 34 (D) 58.2 (E) 58

Day 4 · Questions 61–80

20 questions

Date: _____ Minutes: _____ Score: _____ / 20

🔢 Number sense & arithmetic (continued)

61. What are all the factors of 12?

(A) 12, 24, 36
(B) 1, 2, 4, 6, 12
(C) 12, 24, 36, 48
(D) 1, 2, 3, 4, 6, 12
(E) 2, 3, 4, 6

62. Which of the following numbers has the greatest value?

(A) 1.4378 (B) 1.07548 (C) 1.43592 (D) 0.89409 (E) 1.43785

63. $1\frac{1}{2} + 0.750 + 0.3010 =$ _____.

(A) 1.001 (B) 2.051 (C) 2.055 (D) 2.351 (E) 2.551

64. What percentage of 220 is 26.4?

(A) 909% (B) 99% (C) 40% (D) 27% (E) 12%

65. 0.0515 × 10 is equivalent to _____. ⚠ MISSED TWICE

(A) 5,150 ÷ 100 (B) 5.15 × 10 (C) 0.00515 × 1,000
(D) 510,000 ÷ 10 (E) 5,150 ÷ 10,000

66. 565.684 ÷ 62.3 = _____.

(A) 7.08 (B) 7.68 (C) 9.08 (D) 9.081 (E) 10.08

67. −3 − [(4 − 3) − (3 + 4)] = _____.

(A) 12 (B) 6 (C) 3 (D) − 6 (E) − 9

68. Durant's Trading Company earned profits of $ 75,000 in 1990. In 1998, their profit was $ 4,500,000. The profit from 1998 was how many times as great as it was in 1990?

(A) 2 (B) 4 (C) 6 (D) 10 (E) 60

69. In the multiplication problem shown, ◇ is equal to which of the following digits?

```
  14◇
×   8
------
1,168
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(A) 0 (B) 1 (C) 5 (D) 6 (E) 8

70. What is the second to greatest of the numbers below?

(A) 8.302 (B) 8.203 (C) 8.230 (D) 8.032 (E) 8.320

71. 4,■86

The ■ in the number above represents a digit from 0 through 9. If the number is less than 4,486, what is the greatest possible value for ■?

(A) 0 (B) 3 (C) 4 (D) 7 (E) 9

72. Of the following, which is the closest value to $\frac{998 \times 1{,}004}{48 \times 52}$?

(A) 40 (B) 200 (C) 400 (D) 2,000 (E) 4,000

73. Which of the following lists contains only prime numbers?

(A) 0, 1, 2, 3, 5
(B) 1, 2, 3, 4, 5
(C) 2, 3, 5, 7, 11
(D) 3, 7, 11, 13, 15
(E) 5, 9, 13, 17, 23

📝 Multi-step word problems

- Underline the question. Most traps are a middle step offered as a choice.
- Use every number in the problem — or know why one is a distraction.
- Check your answer back in the story.

74. The Crown Movie Theater collected $705 yesterday. If 54 adults paid $9.50 each and each child paid $3.50 less than each adult, how many children paid admissions?

(A) 32 (B) 38 (C) 29 (D) 25 (E) 42

75. A coat is on sale for $144 after a discount of 20%. Find the original price.

(A) $102.40 (B) $153.60 (C) $160 (D) $180 (E) $148

76. Rachel worked one Saturday from 8:30 a.m. until 3 p.m. at the rate of $4.65 per hour. How much did she receive?

(A) $19.88 (B) $22.53 (C) $19.00 (D) $22.00 (E) $30.23

77. If the outer diameter of a metal pipe is 3.01 inches and the inner diameter is 2.21 inches, the thickness of the metal is _____. ⚠ MISSED TWICE

(A) 0.40 in (B) 0.90 in (C) 1.42 in (D) 1.94 in (E) 2.39 in

78. Johnny picked apples from 9:00 a. m. to 11:30 a. m. and gathered 200 apples. He wants to pick a total of at least 600 apples before 7:15 p. m. If he plans to pick apples at the same rate, what is the latest time that he can start picking apples again?

(A) 1:15 p. m. (B) 1:45 p. m. (C) 2:15 p. m.
(D) 2:45 p. m. (E) 3:15 p. m.

79. A garbage company charges $10 for each of the first 3 months of service and $16 each month after the first 3 months. What is the average cost per month of using this garbage company for the first 12 months?

(A) $11.50 (B) $13.00 (C) $14.50 (D) $14.80 (E) $18.50

80. Sophie is making bows for packages and uses 1.25 meters of ribbon per bow. If she has 12 meters of ribbon and makes as many complete bows as possible, how much ribbon, in meters, will remain?

(A) 0.60 (B) 0.75 (C) 0.85 (D) 1.75 (E) 1.85

Day 5 · Questions 81–99

19 questions

Date: _____ Minutes: _____ Score: _____ / 19

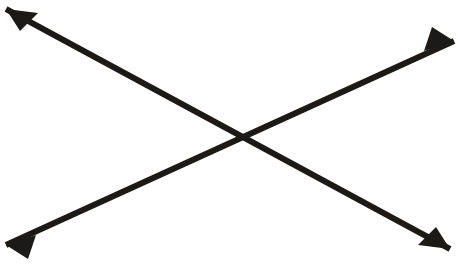
Angles, triangles & shapes

- A triangle's angles add to 180°, a straight line is 180°, a full turn is 360°.
- Look for hidden facts: equal-side marks, right angles, parallel lines.
- "Not drawn to scale" — trust the numbers, not the picture.

81. Each of the equal legs of an isosceles triangle is 3 more than the base, and the perimeter is 30. Find the measure of one of the equal legs.

(A) 12 (B) 10 (C) 7 (D) 11 (E) 15

82. Which of the following statements is true about the two lines below?



(A) The two lines are parallel but not perpendicular.

(B) The two lines are perpendicular but not parallel.

(C) The two lines are both parallel and perpendicular.

(D) The two lines are neither parallel nor perpendicular

(E) It cannot be determined from the information given.

83. Which of the following figures is not a polygon?

(A) Decagon (B) Cone (C) Triangle (D) Rhombus (E) Pentagon

84. The perimeter of a 6-sided polygon is 56 cm. The length of three sides is 9 cm each. The length of two other sides is 8 cm each. What is the length of the missing side?

(A) 11cm

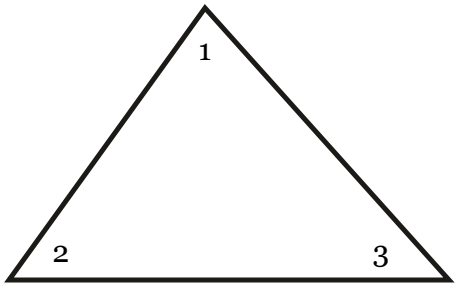
(B) 12 cm

(C) 13 cm

(D) 10 cm

(E) It cannot be determined from the information given.

85. **NOTE:** Figure not drawn to scale.
Which of the following is true?



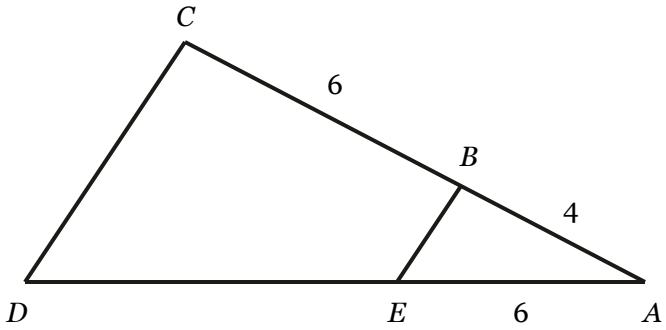
$m\angle 2 = 60°$

(A) $m\angle 1 + m\angle 3 > 180°$
(B) $m\angle 1 + m\angle 3$
(C) $m\angle 1 = m\angle 3$
(D) $m\angle 1 - m\angle 3 > m\angle 2$
(E) $m\angle 1 + m\angle 3 = 120°$

86. How many edges does a heptagonal prism have? (A heptagon is a polygon with 7 sides.)

(A) 6
(B) 12
(C) 14
(D) 18
(E) 21

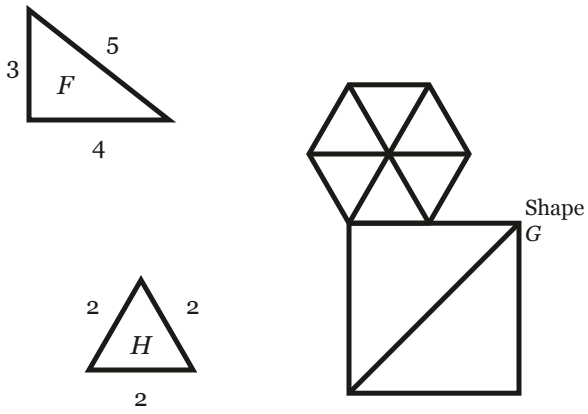
87. Triangles *ABE* and *ACD* are similar. Find the length of *DE*. ⚠ MISSED TWICE



(A) 9 (B) 15 (C) 4 (D) 11 (E) 8

88. Shape *G* is made from pieces *H* and *F*. What is the perimeter of shape *G*?

Note: Figure not drawn to scale.



(A) 22 (B) 23 (C) 24 (D) 27 (E) 29

89. In Figure 2, a cube is divided into two pieces by the plane. What is the shape formed by the intersection of the cube and plane?

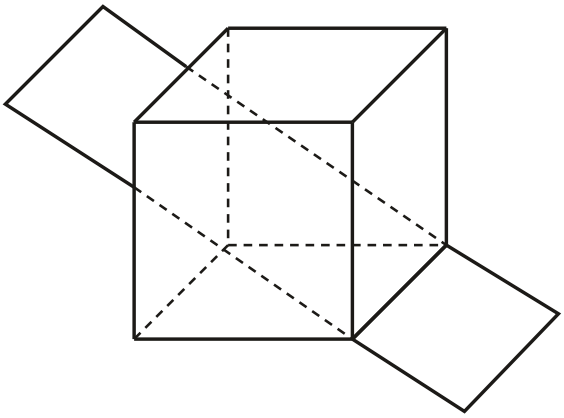
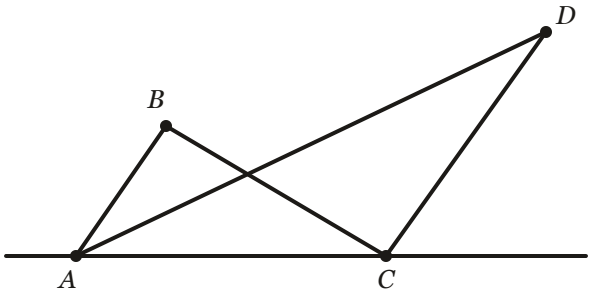


Figure 2

(A) Circle (B) Square (C) Triangle (D) Pentagon (E) Rectangle

90. In the figure below, $\overline{AB}$ and $\overline{CD}$ are parallel. In addition, $m\angle ABC = 100°$, $m\angle ACD = 130°$, and $\overline{AD}$ bisects $\angle BAC$. What is $m\angle D$?

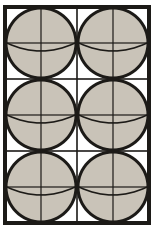


(A) 20° (B) 25° (C) 40° (D) 50° (E) 80°

91. Rafael walks 2 miles east, 2 miles north, 4 miles east, 5 miles north, and 5 miles west. What is the shortest distance between Rafael and his starting point?

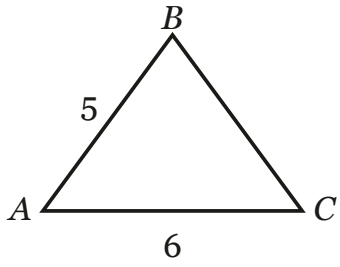
(A) 18 miles
(B) $5\sqrt{2}$ miles
(C) $\sqrt{170}$ miles
(D) $4\sqrt{3}$ miles
(E) 8 miles

92. The top view of a rectangular package of 6 tightly packed balls is shown. If each ball has a radius of 2 centimeters, which of the following are closest to the dimensions, in centimeters, of the rectangular package?



(A) $2 \times 3 \times 6$
(B) $4 \times 6 \times 6$
(C) $2 \times 4 \times 6$
(D) $4 \times 8 \times 12$
(E) $6 \times 8 \times 12$

93. In triangle ABC shown, if $AB = BC$, what is the perimeter of the triangle?



(A) 11 (B) 15 (C) 16 (D) 17 (E) 30

Perimeter, area & volume

- Perimeter = around the edge (add). Area = inside (multiply).
- Radius is half the diameter — check which one you were given.
- Make a length k times bigger and the area grows k × k times.

94. The perimeter of a rectangle measuring 3 × 8 units is equal to the

(A) area of the same rectangle in square units

(B) perimeter of a regular pentagon with side lengths of 3 units

(C) area of a triangle with the base length of 3 units and height of 8 units

(D) area in square units of a rectangle measuring 2 units by 11 units

(E) area in square units of a square measuring 22 units on a side

95. A triangle has vertices located at the points (3, 2), (3, 9), and (7, 2) on the coordinate plane. Find its area.

(A) 10.5 square units
(B) 14.0 square units
(C) 18.0 square units
(D) 28.0 square units
(E) 31.5 square units

96. Determine the ratio of the area of a circle with radius 2 to the area of a circle with radius 4. ⚠ MISSED TWICE

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$ (E) $\frac{2}{5}$

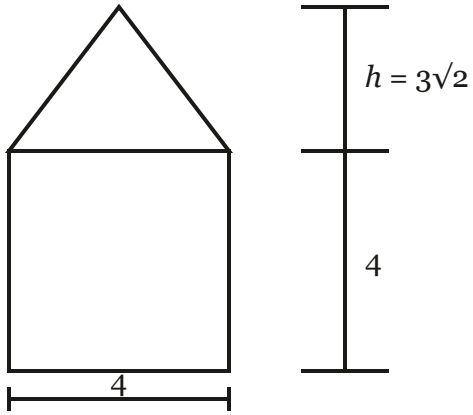
97. If the area of a square is $4a^2$, find its perimeter. ⚠ MISSED TWICE

(A) $6a$ (B) $8a^2$ (C) $12a$ (D) $6a^2$ (E) $8a$

98. Metal screws weigh 3 kilograms per cubic meter. The screws are packaged in a cube whose edge is 1.5 meter. Find the weight of the cube. Round the answer to the nearest tenth of a kilogram.

(A) 9.6 kilograms
(B) 10.125 kilograms
(C) 10.1 kilograms
(D) 9.65 kilograms
(E) 12.5 kilograms

99. Find the total area of the figure.



(A) $4\sqrt{6} + 4$ (B) $3\sqrt{2} + 4$ (C) $5\sqrt{3} + 12$ (D) $16 + 6\sqrt{2}$ (E) $4\sqrt{3} + 2\sqrt{2}$

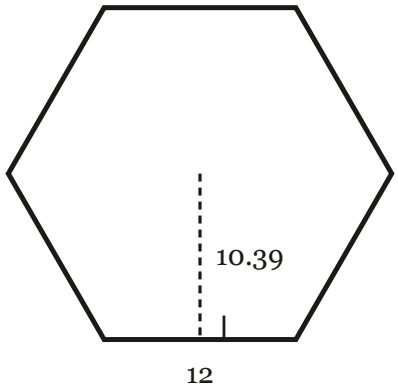
Day 6 · Questions 100–118

19 questions

Date: _____ Minutes: _____ Score: _____ / 19

Perimeter, area & volume (continued)

100. What is the area of the regular hexagon shown below?



(A) 72 (B) 124.68 (C) 374.04 (D) 748.08 (E) 187.02

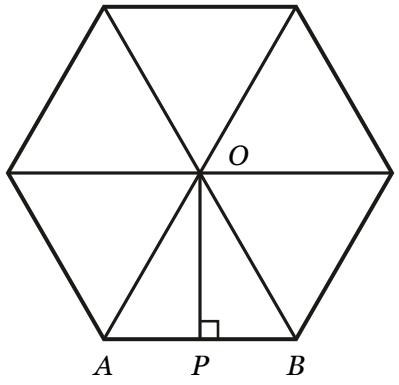
101. The area of a given rectangle is 24 centimeters. If the measure of each side is multiplied by 3, what is the area of the new figure?

(A) 48 cm (B) 72 cm (C) 216 cm (D) 13,824 cm (E) 144 cm

102. What is the distance between (−10, −13) and (−16, −9) along the line connecting them?

(A) 8 (B) 10 (C) $2\sqrt{13}$ (D) $4\sqrt{13}$ (E) 13

103. Calculate the area of the hexagon. $OP = 2\sqrt{3}$, $AB = 4$.

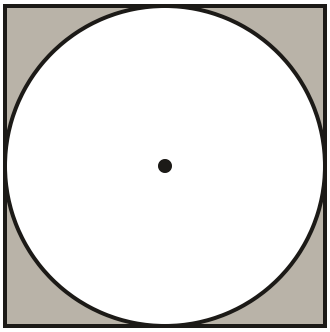


(A) $96\sqrt{3}$ (B) $32\sqrt{3}$ (C) 32 (D) $24\sqrt{3}$ (E) 24

104. From a piece of tin in the shape of a square 6 inches on a side, the largest possible circle is cut out. Of the following, the ratio of the area of the circle to the area of the original square is closest in value to _____. ⚠ MISSED TWICE

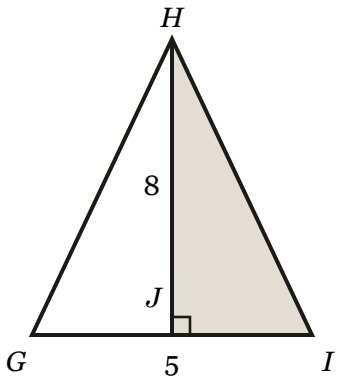
(A) $\frac{4}{5}$ (B) $\frac{2}{3}$ (C) $\frac{3}{5}$ (D) $\frac{7}{9}$ (E) $\frac{1}{2}$

105. If the radius of the circle below is 5 cm, what is the area of the shaded region, in terms of π? *(Note: $A = \pi r^2$)*



(A) $100\pi - 25$ cm^2 (B) $100\pi - 100$ cm^2 (C) $100 - 5\pi$ cm^2
(D) $100 - 25\pi$ cm^2 (E) $25\pi - 9$ cm^2

106. In the figure below, *JH* is the height of the triangle and measures 8 cm, *GI* = 5 cm, and *GH* = *HI*. What is the area of triangle *JHI*? ⚠ MISSED TWICE



(A) 10 cm^2 (B) 13 cm^2 (C) 20 cm^2 (D) 21 cm^2 (E) 40 cm^2

107. Which translation moves a point from (4, −3) to (−2, 5)?

(A) 2 units left, 2 units up
(B) 2 units left, 2 units down
(C) 6 units left, 8 units up
(D) 6 units right, 8 units down
(E) 8 units right, 6 units down

108. Two cardboard boxes have equal volume. The dimensions of one box are 6 × 6 × 10. If the length of the other box is 3 and the width is 10, what is the height of the second box?

(A) 2 (B) 5 (C) 10 (D) 12 (E) 16

109. The length of rectangle A is 6 times the length of rectangle B. The width of rectangle A is $\frac{4}{3}$ the width of rectangle B. If the area of rectangle A is x times the area of rectangle B, what is the value of x?

(A) $\frac{9}{2}$ (B) 6 (C) 8 (D) $\frac{20}{3}$ (E) 12

Charts, averages & probability

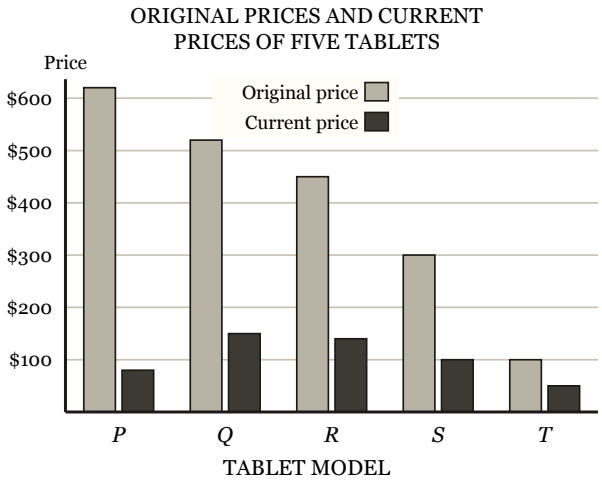
- Average × how many = total. Work with totals.
- Read the chart's scale and key before reading a bar.
- Median: put the numbers in order first.

110. Which of the following pairs of numbers does NOT have an average equal to half the average of 8 and 16?

(A) 2 and 4 (B) 4 and 8 (C) 5 and 7 (D) 6 and 6 (E) 12 and 0

---

111. The tablet model that has the highest current price was originally introduced at a price of



(A) $620 (B) $520 (C) $450 (D) $300 (E) $140

112. The table represents the frequency distribution of students' weights. Find the mean weight and round to the nearest tenth.

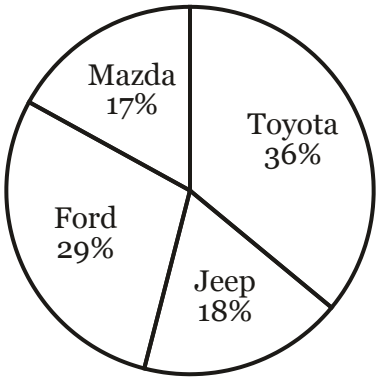
Weights of the students

Pounds, p_i	Frequency, f_i	$p_i f_i$
98	2	
105	1	
109	1	
120	3	
126	2	
132	4	

(A) 120.3 (B) 118.2 (C) 122.1 (D) 116.9 (E) 119.2

113. The car dealership sold 1,800 cars last year. According to the chart, how many Mazdas did the dealership sell? ⚠ MISSED TWICE

Cars Sold by Car Dealership



(A) 648 (B) 306 (C) 324 (D) 522 (E) 486

114. In 2002, what percent of all television shows were detective stories?

Major Network Television Shows

	2002	2003	2004	2005	Summary
Reality	2	3	12	14	31
Comedy	7	9	8	7	31
Law	3	6	7	5	21
Detective	8	9	6	12	35
Cartoons	10	12	14	16	52
News	20	24	21	15	80
Summary	**50**	**63**	**68**	**69**	**250**

(A) 20% (B) 18% (C) 16% (D) 23% (E) 24%

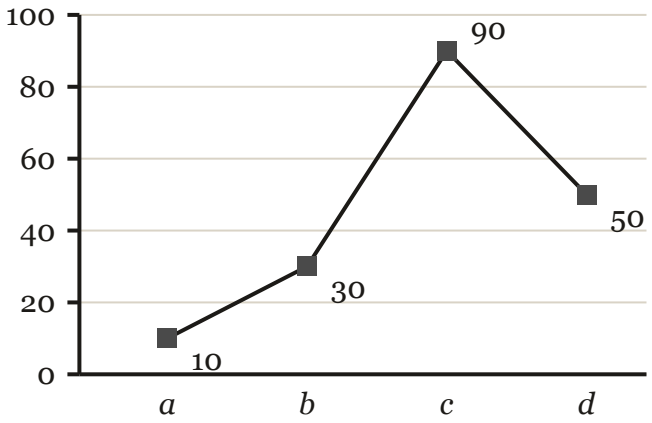
115. The total number of cartoons in the year 2004 and 2005 represent what percent of all shows in the years 2002 through and including 2005?

Major Network Television Shows

	2002	2003	2004	2005	Summary
Reality	2	3	12	14	31
Comedy	7	9	8	7	31
Law	3	6	7	5	21
Detective	8	9	6	12	35
Cartoons	10	12	14	16	52
News	20	24	21	15	80
Summary	**50**	**63**	**68**	**69**	**250**

(A) 18% (B) 24% (C) 22% (D) 12% (E) 16%

116. Examine the graph and then select the correct answer.



(A) $d > a + b + c$ (B) $d = a + b + c$ (C) $d = a + b$

(D) $c - a < b + c$ (E) $a + c < b + d$

117. Karen gets paid a weekly salary and a commission for every sale that she makes. The table below shows the number of sales and her pay for different weeks. Which of the following equations represents Karen's weekly pay? ⚠ MISSED TWICE

Karen's sales and her pay for different weeks

Sales	2	7	4	8
Pay	$380	$580	$460	$620

(A) $y = 90x + 200$ (B) $y = 90x - 200$ (C) $y = 40x - 200$

(D) $y = 90x + 400$ (E) $y = 40x + 300$

118. Which measure for the center of a small sample set is most affected by outliers?

⚠ MISSED TWICE

(A) Mean

(B) Median

(C) Mode

(D) None of the above

(E) All of the above

Day 7 · Questions 119–137

19 questions

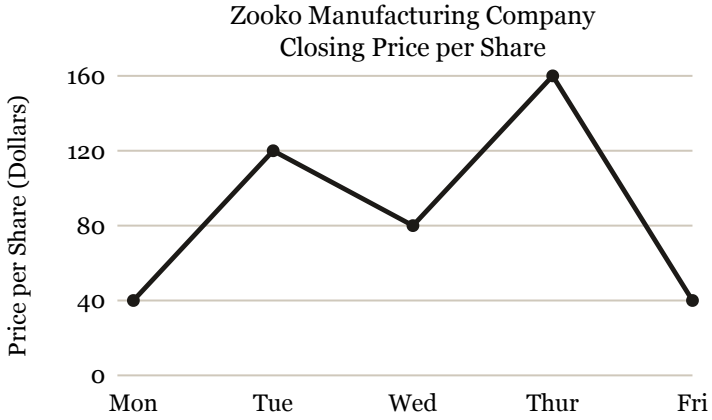
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Charts, averages & probability (continued)

119. Dwayne has received the following scores on his math tests: 78, 92, 83, 97. What score must Dwayne get on his next math test to have an overall average of at least 90?

(A) 89 (B) 98 (C) 95 (D) 99 (E) 100

120. Mike bought 10 shares of Zooko stock at the closing price on Tuesday and sold them at the closing on Friday. How much money did Mike lose on his investment?



(A) $ 80 (B) $ 200 (C) $ 800 (D) $ 2,000 (E) $ 95

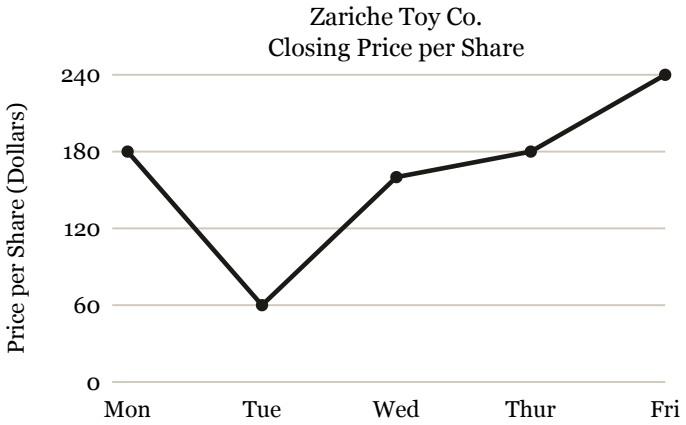
121. A number cube with the faces numbered 1 through 6 is rolled three times. What is the probability that all three rolls will result in the number one?

(A) $\frac{1}{2}$ (B) $\frac{1}{6}$ (C) $\frac{3}{216}$ (D) $\frac{1}{120}$ (E) $\frac{1}{216}$

122. The average of three numbers is 5. What is three times the sum of the three numbers?

(A) 6 (B) 15 (C) 30 (D) 45 (E) 90

123. Josie bought 10 shares of Zariche stock at the closing price Monday and sold them at the closing price on Friday. What was Josie's profit on this investment?



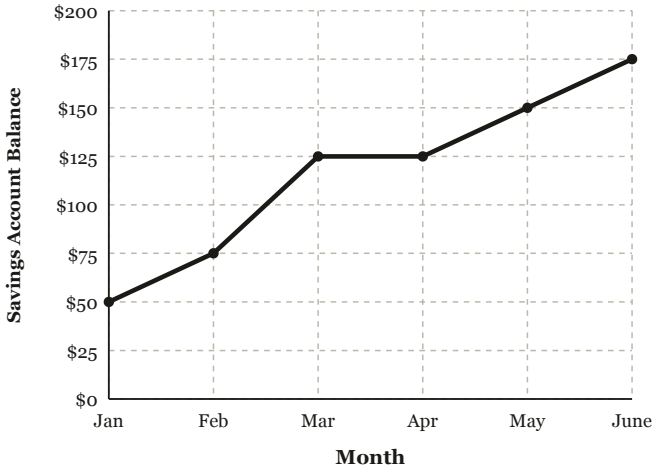
(A) $ 60
(B) $ 96
(C) $600
(D) $960
(E) None of the above

124. If a Set R contains four positive integers whose average is 7, what is the greatest possible value of one of the integers?

(A) 4 (B) 9 (C) 25 (D) 33 (E) 36

125. The graph below shows the balance of a savings account, which was opened in January, over a 6 month period.

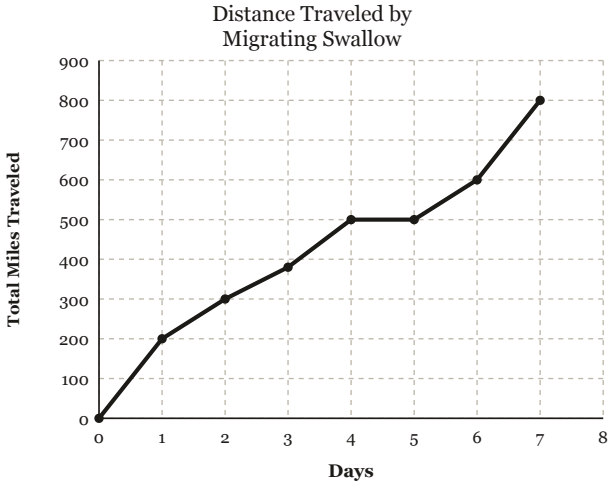
The savings account balance in June was how much greater than in January?



(A) $25 (B) $50 (C) $100 (D) $125 (E) $175

126. The graph below shows the distance travelled by a migrating swallow to nesting grounds 800 miles away.

At the beginning of day 2, the swallow had travelled 200 miles. What total distance did the swallow travel from the beginning of day 2 through the end of day 4?



(A) 200 miles (B) 300 miles (C) 400 miles (D) 500 miles (E) 600 miles

127. 30, −15, 10, 35, −15, −30, 20
What is the median of the list above?

(A) −15 (B) 5 (C) 10 (D) 20 (E) 35

Number properties & logic

- "MUST be true" — one counter-example kills a choice. Try 0, 1, a negative, a fraction.
- For "to be sure of…", picture the unluckiest case, then add 1.
- For a made-up operation, copy the rule exactly and plug in.

128. If a is a negative integer and b is a positive one, which of the following statements is true?

(A) $ab < ba$ (B) $a + b > b - a$ (C) $a^2 < 0$
(D) $b^2 < 0$ (E) $ba \geq ab$

129. If a and b are integers and $a > 0$ and $b < 0$, select the statement that is true for all cases of a and b.

(A) $ab > 0$ (B) $ab \leq 0$ (C) $a^2 > b^2$ (D) $a^3 < b^3$ (E) $2a \leq b$

130. What is the 42nd item in the pattern:

▲ ○○□ ▲ ○○□ ▲ …

(A) ○ (B) ▲
(C) □ (D) None of the above
(E) all of the above

131. What is the probability of randomly picking the winner and runner-up from a race of 4 horses and distinguishing which is the winner? ⚠ MISSED TWICE

(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{1}{16}$ (D) $\frac{1}{3}$ (E) $\frac{1}{12}$

132. How many socks would you need to remove from a drawer containing 10 blue socks, 15 black socks, and 5 red socks, to insure that you have at least 2 matching socks? ⚠ MISSED TWICE

(A) 3 (B) 4 (C) 16 (D) 17 (E) 28

133. If 6 is a factor of a certain number, what must also be factors of that number?

(A) 1, 2, 3, and 5 (B) 2, and 3 only (C) 6 only
(D) 2 and 5 only (E) 1, 2, and 3

134. In the fraction $\frac{1}{\Delta - 3}$, Δ can be replaced by all of the following EXCEPT _____.

(A) $+3$ (B) $+2$ (C) 0 (D) -1 (E) -2

135. Pictures for a magazine layout are numbered consecutively starting with 10. They are arranged in numerical order, three to a page, from left to right. Which of the following could be the number of a picture in the middle of a page?

(A) 40 (B) 43 (C) 46 (D) 53 (E) 64

136. On a certain obstacle course involving 20 participants, 12 scored more than 50 points and 8 scored at least 52 points. How many participants scored exactly 51 points?

(A) 4
(B) 5
(C) 6
(D) 7
(E) It cannot be determined from the information given.

137. Tricia shelves consecutively numbered books at the library starting with 1. They are placed 8 to a shelf in numerical order. Which of the following numbers would NOT be the third book on a shelf?

(A) 3 (B) 19 (C) 23 (D) 67 (E) 83